

Introduction to model-based community analysis

Bert van der Veen

University of Bayreuth

Gathering data

We go out, register species at multiple sites



Figure 1: Geir-Harald Strand / NIBIO

Gathering data

We go out, register species at multiple sites



Figure 1: Geir-Harald Strand / NIBIO

What does community data look like?

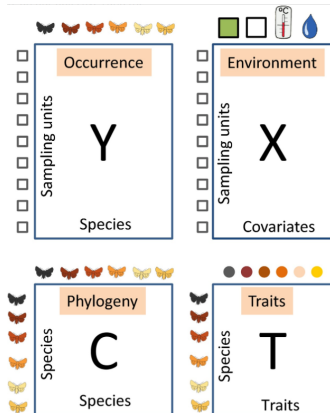
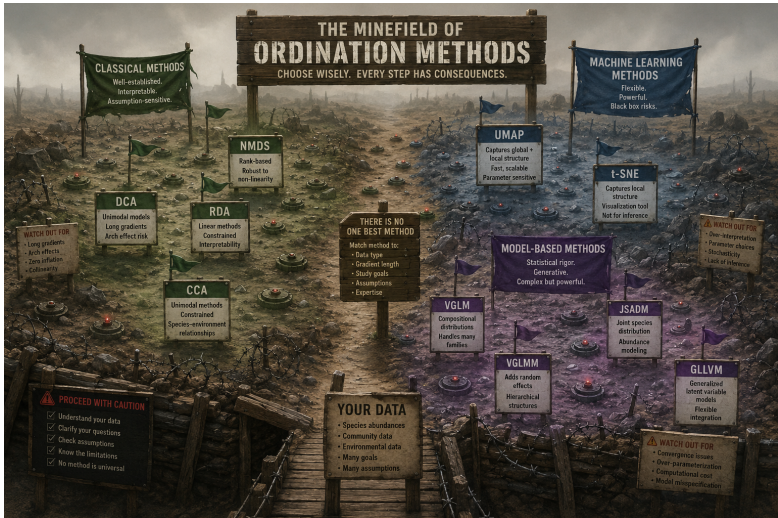


Figure 2: Ovaskainen et al. (2017) Fig. 3

Now pick an analysis



Outline of the workshop

Day	Topics
1	Vector GLMs → Vector GLMMs → Model checking
2	Hierarchical responses → JSDMs → Diversity
3	Model-based ordination → Constrained/concurrent → Conditioning
4	Unimodal responses → Spatial/temporal → Own data

This lecture: why model-based methods, properties of community data, and how models address them.

From VGLMs and JSDMs to ordination and beyond

- ▶ **VGLMs**: multispecies regression; species-specific responses to the environment
- ▶ **JSDMs**: add residual species correlations; why do species co-occur after accounting for the environment?
- ▶ **Ordination**: latent variables become the primary object; what are the main unmeasured gradients driving the community?

From VGLMs and JSDMs to ordination and beyond

- ▶ **VGLMs:** multispecies regression; species-specific responses to the environment
- ▶ **JSDMs:** add residual species correlations; why do species co-occur after accounting for the environment?
- ▶ **Ordination:** latent variables become the primary object; what are the main unmeasured gradients driving the community?

Each step builds on the previous. All use the same statistical framework (GLLVMs).

Community ecology

We collect data on many species at many sites. We want to understand:

- ▶ Which species co-occur, and why?
- ▶ How do species respond to the environment?
- ▶ How do communities change across gradients?
- ▶ What drives diversity?

Community ecology

We collect data on many species at many sites. We want to understand:

- ▶ Which species co-occur, and why?
- ▶ How do species respond to the environment?
- ▶ How do communities change across gradients?
- ▶ What drives diversity?

Classical ordination methods (NMDS, CCA, RDA, PCA, PCoA, ...) have been the dominant tools for decades.

Community ecology

We collect data on many species at many sites. We want to understand:

- ▶ Which species co-occur, and why?
- ▶ How do species respond to the environment?
- ▶ How do communities change across gradients?
- ▶ What drives diversity?

Classical ordination methods (NMDS, CCA, RDA, PCA, PCoA, ...) have been the dominant tools for decades.

They have serious limitations.

You may already know these methods

Classical methods are well-established and widely expected:

- ▶ Many reviewers and journals still expect NMDS, CCA, PERMANOVA
- ▶ Textbooks and courses often teach only classical methods

You may already know these methods

Classical methods are well-established and widely expected:

- ▶ Many reviewers and journals still expect NMDS, CCA, PERMANOVA
- ▶ Textbooks and courses often teach only classical methods

In simple cases (short gradients, balanced designs, no hierarchical structure) they are reasonable approximations.

You may already know these methods

Classical methods are well-established and widely expected:

- ▶ Many reviewers and journals still expect NMDS, CCA, PERMANOVA
- ▶ Textbooks and courses often teach only classical methods

In simple cases (short gradients, balanced designs, no hierarchical structure) they are reasonable approximations.

When they fail: long gradients, sparse count data, hierarchical study designs, prediction under new conditions.

You may already know these methods

Classical methods are well-established and widely expected:

- ▶ Many reviewers and journals still expect NMDS, CCA, PERMANOVA
- ▶ Textbooks and courses often teach only classical methods

In simple cases (short gradients, balanced designs, no hierarchical structure) they are reasonable approximations.

When they fail: long gradients, sparse count data, hierarchical study designs, prediction under new conditions.

This workshop teaches the framework that makes the approximation explicit and testable.

`gllvm`, `Hmsc`, and `mvabund` are now standard tools in the community ecology literature.

A strategy by necessity

Community ecology needed tools for multispecies data long before the statistical machinery existed to model it properly.

- ▶ Distance matrices and transformations were the best available options in the 1970s–90s
- ▶ They became standard practice: embedded in textbooks, software, and reviewer expectations
- ▶ Not a principled choice, but a practical one: there was no better alternative

A strategy by necessity

Community ecology needed tools for multispecies data long before the statistical machinery existed to model it properly.

- ▶ Distance matrices and transformations were the best available options in the 1970s–90s
- ▶ They became standard practice: embedded in textbooks, software, and reviewer expectations
- ▶ Not a principled choice, but a practical one: there was no better alternative

The field built its analytical culture around these tools.

A strategy by necessity

Community ecology needed tools for multispecies data long before the statistical machinery existed to model it properly.

- ▶ Distance matrices and transformations were the best available options in the 1970s–90s
- ▶ They became standard practice: embedded in textbooks, software, and reviewer expectations
- ▶ Not a principled choice, but a practical one: there was no better alternative

The field built its analytical culture around these tools.

Now there is a principled alternative.

What do classical methods assume?

Method	Purpose	Assumptions
PCA	Unconstrained	Gaussian; linear; constant variance
RDA	Constrained	Gaussian; linear; constant variance
CA	Unconstrained	Approximate species packing
CCA	Constrained	Approximate species packing
DCA	Unconstrained	Approximate species packing; rescaled
NMDS	Unconstrained	Distance metric reflects ecological similarity
db-RDA	Constrained visualisation	Distance metric reflects ecological similarity
PERMANOVA	Hypothesis test	Homogeneous within-group dispersion

What do classical methods assume?

Method	Purpose	Assumptions
PCA	Unconstrained	Gaussian; linear; constant variance
RDA	Constrained	Gaussian; linear; constant variance
CA	Unconstrained	Approximate species packing
CCA	Constrained	Approximate species packing
DCA	Unconstrained	Approximate species packing; rescaled
NMDS	Unconstrained	Distance metric reflects ecological similarity
db-RDA	Constrained visualisation	Distance metric reflects ecological similarity
PERMANOVA	Hypothesis test	Homogeneous within-group dispersion

A single process-oriented framework can address all of these tasks.
 We will return to this throughout the workshop.

What do classical methods do well?

These methods have been the backbone of community ecology for 50+ years, and for good reason:

- ▶ **Easy to use:** `vegan` is well-documented; functions are fast and accessible
- ▶ **Known behavior:** artifacts and failure modes are well-characterised; large published literature
- ▶ **Computationally fast:** eigenvalue decomposition scales well; permutation is parallelisable
- ▶ **Permutation inference:** PERMANOVA and ANOSIM require no distributional assumptions for the test statistic
- ▶ **Variance partitioning:** straightforward with `vegan::varpart()`
- ▶ **Widely understood:** biplots and triplots are familiar to most referees and readers

What do classical methods do well?

These methods have been the backbone of community ecology for 50+ years, and for good reason:

- ▶ **Easy to use:** `vegan` is well-documented; functions are fast and accessible
- ▶ **Known behavior:** artifacts and failure modes are well-characterised; large published literature
- ▶ **Computationally fast:** eigenvalue decomposition scales well; permutation is parallelisable
- ▶ **Permutation inference:** PERMANOVA and ANOSIM require no distributional assumptions for the test statistic
- ▶ **Variance partitioning:** straightforward with `vegan::varpart()`
- ▶ **Widely understood:** biplots and triplots are familiar to most referees and readers

They work best when gradients are short, data are not too sparse, and exploration rather than inference or prediction is the goal.

Pattern recognition or process modelling?

Classical methods excel at exploration and visualisation: they reveal *what* the community looks like.

Pattern recognition or process modelling?

Classical methods excel at exploration and visualisation: they reveal *what* the community looks like.

Ecology asks a different question: *why* does the community look that way?

- ▶ Why do species co-occur after accounting for the environment?
- ▶ How do communities respond to change?
- ▶ What processes generate the diversity we observe?

Pattern recognition or process modelling?

Classical methods excel at exploration and visualisation: they reveal *what* the community looks like.

Ecology asks a different question: *why* does the community look that way?

- ▶ Why do species co-occur after accounting for the environment?
- ▶ How do communities respond to change?
- ▶ What processes generate the diversity we observe?

Pattern recognition cannot answer process questions. A model for the data-generating process can.

Classical methods: no random effects

Classical ordination cannot accommodate hierarchical study designs:

- ▶ Repeated measurements of sites
- ▶ Plots nested in sites
- ▶ Spatial or temporal autocorrelation

Classical methods: no random effects

Classical ordination cannot accommodate hierarchical study designs:

- ▶ Repeated measurements of sites
- ▶ Plots nested in sites
- ▶ Spatial or temporal autocorrelation

Adding a random effect to CCA or NMDS is not possible.

In model-based analysis, mixed effects are a natural extension of the model, the same tools you use in a GLMM.

Classical methods: no uncertainty

NMDS and CA ordination scores have no standard errors.

Classical methods: no uncertainty

NMDS and CA ordination scores have no standard errors.

- ▶ You cannot test whether a species' position on the ordination axis is significantly different from zero
- ▶ You cannot propagate uncertainty in ordination scores to downstream analyses
- ▶ Biplots give the impression of precision that is not there

Classical methods: no uncertainty

NMDS and CA ordination scores have no standard errors.

- ▶ You cannot test whether a species' position on the ordination axis is significantly different from zero
- ▶ You cannot propagate uncertainty in ordination scores to downstream analyses
- ▶ Biplots give the impression of precision that is not there

Model-based methods inherit full uncertainty quantification from the statistical model.

Classical methods: no prediction

Distance-based methods do not define a model for the data.

Classical methods: no prediction

Distance-based methods do not define a model for the data.

- ▶ NMDS cannot predict the expected occurrence of a species at a new site
- ▶ CCA cannot forecast community composition under future climate
- ▶ Species richness cannot be derived from an NMDS ordination

Classical methods: no prediction

Distance-based methods do not define a model for the data.

- ▶ NMDS cannot predict the expected occurrence of a species at a new site
- ▶ CCA cannot forecast community composition under future climate
- ▶ Species richness cannot be derived from an NMDS ordination

Model-based methods define a data-generating process.

Predictions for new sites, new conditions, and derived quantities follow from the model.

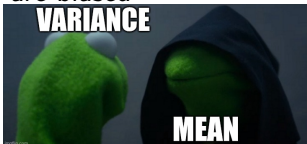
Classical methods: transformation or distances

Community data is sparse, overdispersed, and non-normal.

Classical methods apply one of two workarounds:

Data transformation

- ▶ $\log(y + 1)$, square root, presence/absence
- ▶ Does not remove the mean-variance trend (O'Hara & Kotze 2010)
- ▶ Variance not stabilized; back-transformed estimates are biased



Distance matrix

- ▶ Bray-Curtis, Jaccard, Euclidean, ...
- ▶ Discards raw data structure
- ▶ Mean-variance assumptions become implicit in the metric (Warton et al. 2012)

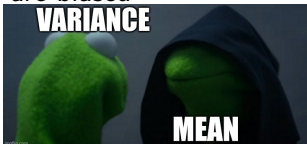
Classical methods: transformation or distances

Community data is sparse, overdispersed, and non-normal.

Classical methods apply one of two workarounds:

Data transformation

- ▶ $\log(y + 1)$, square root, presence/absence
- ▶ Does not remove the mean-variance trend (O'Hara & Kotze 2010)
- ▶ Variance not stabilized; back-transformed estimates are biased



Distance matrix

- ▶ Bray-Curtis, Jaccard, Euclidean, ...
- ▶ Discards raw data structure
- ▶ Mean-variance assumptions become implicit in the metric (Warton et al. 2012)

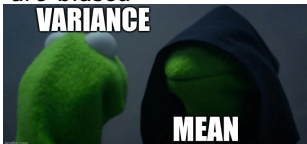
Classical methods: transformation or distances

Community data is sparse, overdispersed, and non-normal.

Classical methods apply one of two workarounds:

Data transformation

- ▶ $\log(y + 1)$, square root, presence/absence
- ▶ Does not remove the mean-variance trend (O'Hara & Kotze 2010)
- ▶ Variance not stabilized; back-transformed estimates are biased

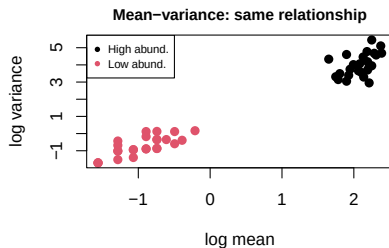
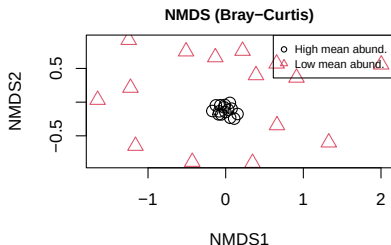


Distance matrix

- ▶ Bray-Curtis, Jaccard, Euclidean, ...
- ▶ Discards raw data structure
- ▶ Mean-variance assumptions become implicit in the metric (Warton et al. 2012)

Distance-based methods confound location and dispersion

Warton et al. (2012): equal dispersion parameter, different means.
Bray-Curtis ordination still suggests a dispersion effect.



Low-abundance group appears more dispersed in NMDS, but the dispersion parameter is identical; variance just tracks the mean. **This is a Bray-Curtis artefact, not a real dispersion effect.**

The model-based alternative

Two papers that directly motivated this workshop:



Forum Open Access

The central role of mean-variance relationships in the analysis of multivariate abundance data: a response to Roberts (2017)

David I. Warton Francis K. C. Hui

First published: 26 July 2017 | <https://doi.org/10.1111/2041-210X.12843> | Citations: 28

Warton & Hui (2017): mean-variance relationships are central; distance-based analyses get them wrong

Plant Ecol (2015) 216:669–682
DOI 10.1007/s11258-014-0366-3

Model-based thinking for community ecology

David I. Warton · Scott D. Foster ·
Glenn De'ath · Jakub Stoklosa · Piers K. Dunstan

Warton et al. (2015): a framework for model-based thinking in community ecology

Contemporary model-based methods

In classical ecology, ordination *is* the analysis. In the GLLVM framework, ordination is the tool of choice when it is not possible to estimate species-specific parameters.

Contemporary model-based methods

In classical ecology, ordination *is* the analysis. In the GLLVM framework, ordination is the tool of choice when it is not possible to estimate species-specific parameters.

- ▶ **VGLMs / VGLMMs**: full rank; one parameter per species per covariate
- ▶ **JSDMs**: incorporate species correlations via latent variables
- ▶ **Ordination**: dimension reduction; effects summarised through latent variables

Contemporary model-based methods

In classical ecology, ordination *is* the analysis. In the GLLVM framework, ordination is the tool of choice when it is not possible to estimate species-specific parameters.

- ▶ **VGLMs / VGLMMs**: full rank; one parameter per species per covariate
- ▶ **JSDMs**: incorporate species correlations via latent variables
- ▶ **Ordination**: dimension reduction; effects summarised through latent variables

We begin with full-rank models on Days 1–2 because they make fewer assumptions. Ordination is introduced on Day 3, once the foundations are in place.

All three can be fitted with the `gllvm` R package.

Models are not a panacea

- ▶ **Asymptotic results:** inference relies on large-sample approximations; standard errors and p-values can be unreliable for small datasets or sparse community matrices
- ▶ **Likelihood approximations:** `gllvm` uses a variational approximation; it is fast but not exact, and quality varies with model complexity and data sparsity
- ▶ **Computationally intensive:** complex models with many species and latent variables are slow; Bayesian alternatives (e.g. `Hmsc`) are slower still
- ▶ **Steep learning curve:** model specification, convergence diagnosis, and interpretation require more statistical knowledge than running `vegan`

Models are not a panacea

- ▶ **Asymptotic results:** inference relies on large-sample approximations; standard errors and p-values can be unreliable for small datasets or sparse community matrices
- ▶ **Likelihood approximations:** `gllvm` uses a variational approximation; it is fast but not exact, and quality varies with model complexity and data sparsity
- ▶ **Computationally intensive:** complex models with many species and latent variables are slow; Bayesian alternatives (e.g. `Hmsc`) are slower still
- ▶ **Steep learning curve:** model specification, convergence diagnosis, and interpretation require more statistical knowledge than running `vegan`

The goal is not to replace judgment with software, but to make

Two processes, one model

Our data results from two distinct processes:

The ecological process

- ▶ Species-environment relationships
- ▶ Co-occurrence and interactions
- ▶ Diversity and community assembly

Our primary interest.

The sampling process

- ▶ Hierarchical study design
- ▶ Preferential or opportunistic sampling
- ▶ Imperfect detection
A source of noise.

Two processes, one model

Our data results from two distinct processes:

The ecological process

- ▶ Species-environment relationships
- ▶ Co-occurrence and interactions
- ▶ Diversity and community assembly

Our primary interest.

The sampling process

- ▶ Hierarchical study design
- ▶ Preferential or opportunistic sampling
- ▶ Imperfect detection

A source of noise.

Classical methods: sampling accommodated through study design alone.

Model-based: random effects for hierarchical designs, offsets for effort, detection submodels; the model reflects how the data was collected.

Joint Species Abundance Distribution Models

After accounting for the environment, species still co-occur more (or less) than expected:

- ▶ Unmeasured gradients affect multiple species simultaneously
- ▶ Biotic interactions leave a signature in co-occurrence

Joint Species Abundance Distribution Models

After accounting for the environment, species still co-occur more (or less) than expected:

- ▶ Unmeasured gradients affect multiple species simultaneously
- ▶ Biotic interactions leave a signature in co-occurrence

Classical methods treat co-occurrence as a pattern to visualise. JSADMs model it as a process: residual species correlations, estimated jointly with species-environment relationships.

Joint Species Abundance Distribution Models

After accounting for the environment, species still co-occur more (or less) than expected:

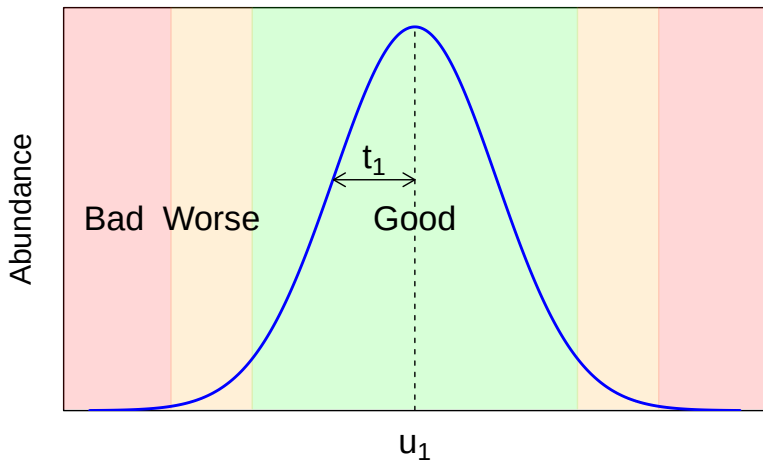
- ▶ Unmeasured gradients affect multiple species simultaneously
- ▶ Biotic interactions leave a signature in co-occurrence

Classical methods treat co-occurrence as a pattern to visualise. JSADMs model it as a process: residual species correlations, estimated jointly with species-environment relationships.

The latent variables in a JSADM represent unmeasured structure in the ecological process: not distances, not ordination scores, but model parameters with standard errors.

Shelford's law makes community data sparse

Ordination can represent sparse data in only a few dimensions.



What ordination brings

When data are too sparse to estimate species-specific parameters, ordination summarises the main gradients via latent variables:

- ▶ Response-scale dimension reduction: no transformation, no distance metric
- ▶ Uncertainty in ordination scores propagates to all downstream inference
- ▶ Predictions for new sites and derived quantities follow from the model

What ordination brings

When data are too sparse to estimate species-specific parameters, ordination summarises the main gradients via latent variables:

- ▶ Response-scale dimension reduction: no transformation, no distance metric
- ▶ Uncertainty in ordination scores propagates to all downstream inference
- ▶ Predictions for new sites and derived quantities follow from the model

The cost: rank constraints on species responses induce bias. The number of latent dimensions must be chosen, and the axes require interpretation. Use ordination when the data cannot support a full-rank model.

Prediction matters for community ecology

A model for the data opens up a coherent framework for forecasting:

- ▶ **Species:** occurrence probability or abundance at unvisited sites or future conditions
- ▶ **Communities:** composition, turnover, and beta diversity across gradients
- ▶ **Diversity:** richness as the sum of occurrence probabilities; no separate estimator needed

Prediction matters for community ecology

A model for the data opens up a coherent framework for forecasting:

- ▶ **Species:** occurrence probability or abundance at unvisited sites or future conditions
- ▶ **Communities:** composition, turnover, and beta diversity across gradients
- ▶ **Diversity:** richness as the sum of occurrence probabilities; no separate estimator needed

Non-parametric estimators work around the absence of a model. When a model is available, richness, diversity indices, and uncertainty follow directly.

Prediction matters for community ecology

A model for the data opens up a coherent framework for forecasting:

- ▶ **Species:** occurrence probability or abundance at unvisited sites or future conditions
- ▶ **Communities:** composition, turnover, and beta diversity across gradients
- ▶ **Diversity:** richness as the sum of occurrence probabilities; no separate estimator needed

Non-parametric estimators work around the absence of a model. When a model is available, richness, diversity indices, and uncertainty follow directly.

The same model used for ordination can forecast diversity under climate change.

Summary: classical approach vs. GLLVM

Problem	Classical approach	GLLVM
Count/binary data	$\log(y + 1)$ transformation	Poisson, NB, or binomial distribution
Long gradient (arch)	DCA rescaling (heuristic)	Unimodal response model
Nested / repeated design	None	Random effects
Species correlation	Ignored	JSDM / residual covariance
No environment measured	NMDS	Unconstrained ordination
Uncertainty	None	Standard errors and CIs

Summary: classical approach vs. GLLVM

Problem	Classical approach	GLLVM
Count/binary data	$\log(y + 1)$ transformation	Poisson, NB, or binomial distribution
Long gradient (arch)	DCA rescaling (heuristic)	Unimodal response model
Nested / repeated design	None	Random effects
Species correlation	Ignored	JSDM / residual covariance
No environment measured	NMDS	Unconstrained ordination
Uncertainty	None	Standard errors and CIs

We adjust the model, not the data.

Summary: classical approach vs. GLLVM

Problem	Classical approach	GLLVM
Count/binary data	$\log(y + 1)$ transformation	Poisson, NB, or binomial distribution
Long gradient (arch)	DCA rescaling (heuristic)	Unimodal response model
Nested / repeated design	None	Random effects
Species correlation	Ignored	JSDM / residual covariance
No environment measured	NMDS	Unconstrained ordination
Uncertainty	None	Standard errors and CIs

We adjust the model, not the data.

Models will be more honest to you than transformations.